

Inscription and Ancient Script Recognition Using OCR

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ABSTRACT

Ancient scripts & inscriptions often offer vital and important understandings to the preceding societies, and separate them effectively that is essential for conservation of cultural remains, decrypting scripts, and protecting the cultural heritage. Usual research generally pay attention on identification of ancient script data in a secluded way, like exemption of old age inscriptions like hand-written scripts. Nonetheless, ancient inscriptions and scripts derive in several arrangements across different mediums, which then creates various challenges for Optical Character Recognition (OCR). Things like erosion, disruption, and inscriptional fragmentation may mis-perceive the process of recognition of ancient inscriptions engravings. To handle this issue, we suggest an ancient inscription recognition framework that can handle different inputs such as stone inscriptions and handwritten scripts. Offering a proportional analysis of various OCR tools & techniques for evaluate how much these are effective and operative in terms of accuracy, recognition of characters, and its processing speed. It plays an important role in inscriptional studies, that consists the inscriptional studies that are crafted on various materials like stones, metal and pottery. This is mostly essential for preserving and digitalizing ancient scripts.

Keywords: Optical Character Recognition (OCR), Script Extraction, Information Retrieval, Convolutional Neural Networks (CNNs), OpenCV (Computer Vision) python library, Image Processing.

Introduction

The study of ancient scripts & inscriptions means one of the most significant markers of primordial, language- based, and epigraphical study. Since the epigraphs of ancient India to the Egyptian ideograms and the Ethiopian scripts of Ge'ez, where each inscriptions inscribed form demonstrates a remarkable

and kept archive of cultural knowledge, traditions, and early organizational systems. Due to various factors like environmental degradation or human activities, the details which ancient inscriptions holds could be lost despite their value. The understanding of old age inscriptions therefore, it stands as an industrial and intellectual challenge.[1],[2]. In context, Optical

Character Recognition (OCR), integrated with Machine Learning (ML) and Deep Neural Networks (DNNs), which offers means to recover fragmented, holding unclear details and documented known information & data from old-image images and bodily remains that might persist non readable.

Challenges in Ancient Script Recognition

Various modern text recognition systems, old-age script recognition offers various multi-dimensional tasks. The heterogeneousness of materials - stones, clay, metallic, palm leaves etc. offers different textures and lighting conditions that could change visual [8] characteristics.

Erosion and surface damage can lead to missing or incomplete characters, whereas handwritten variability and script evolution complicate segmentation and classification multilingual and cross-script characteristics, like hybrid scripts holding Sanskrit, Prakrit, and early Tamil Brahmi, demand adaptive recognition models capable of cross-lingual generalization.[6]

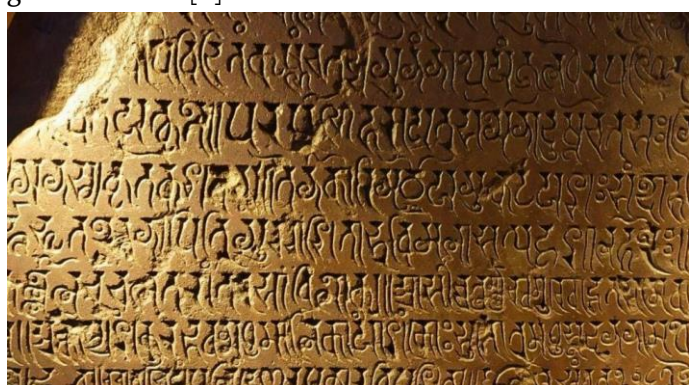


Fig.1. Ancient Sanskrit Inscription show Ancient India influence

Studies have shown that even after pre-processing, conventional OCR accuracy hardly exceeds 70–75% on inscription datasets (Agrawal et al., 2024; Wang et al., 2025). Furthermore, utmost bequest OCR systems miss the mark in character segmentation, mainly for ancient scripts carved in associated strokes or continuous reliefs. Therefore, researchers have moved towards data-driven deep learning models that

acquire and learn discriminative features straight from substantial volumes of interpreted data.

Evolution of OCR and Machine Learning Approaches

The transition from traditional OCR to deep learning-based systems corresponding to the development of computer vision itself. Initial OCR engines in the 1990s, includes the first versions of **Tesseract**, relied on handmade structures like edge histograms and partitioning. These were limited in their capability to generalize across non-standard fonts or degraded images. The overview of **CNN-based models**—notably **LeNet**, **AlexNet**, and **ResNet**—marked a turning point. CNNs, through encrusted convolutional screens, automatically learned tiered illustrations of image features without manual intrusion [7],[8]. Though, CNNs solely couldn't capture the **sequential and contextual relationships** among characters—an essential component for script interpretation and understanding. To address this, **LSTM** networks were presented to the OCR pipelines, establishing the hybrid learning based **CNN- LSTM architectures**. [9]

Significance of the Study

The recognition of early age scripts encompasses far-off past technical innovation—it carries profound insinuations for **cultural, preservation, linguistic research and historical reconstruction**.

Automated OCR-based transcription can exponentially fast-track the progression by altering visual artifacts into machine-readable text, which can be suitable for language-based modelling, ancient database creation, and digital archiving.

In applied standings, such system assistances in not only academically but also heritage institutions, museums, and archives. By digitization of inscriptions & scripts, institutions can preserve delicate pieces & artifacts deprived of any physical handling, which aids in dipping the peril of degradation.[10] The usage of **explainable AI (XAI)** in the recognition progression adds another layer of transparency.

Dataset Limitations and Synthetic Data Generation

One of the utmost tenacious encounters in ancient inscriptions recognition is **data scarcity**. Widely

obtainable datasets are partial in both size and diversity. For instance, the Hwaem Sutra dataset (Jain & Ahn, 2025) comprises about 3,800 samples, whereas most Brahmi and Ge'ez corpora persist underneath 10,000 images. This is inadequate for training high-capacity deep learning networks, which typically necessitate millions of instances for optimal performance. Furthermore, **synthetic data generation** uses **StyleGAN** or **Diffusion Models** and creates pseudo-inscriptions that mimic the graphic variety of authentic relics.[13] This extended dataset empowers robust & strong training and province adaptation, ensuring that representations can perform efficiently even on earlier un-observed scripts writings.

Societal and Interdisciplinary Impact

The implications of this study stretch into education, heritage management, and global cultural equity. Digitalized and recorded calligraphies democratize access to knowledge, allowing apprentices, investigators, and the communal to discover ancient works disadvantaged of environmental or language barriers. Designed for groups go-getting to resuscitate scarce dialects, automated OCR tools propose an vital resource for recuperating missing vocabulary and grammar patterns. The inter-disciplinary mode of this work also opens-up collaborations across various fields. Philologists can employ documented text for proportional grammar revisions, whereas historiographers can execute statistical analyses of time-based and spatial inscription patterns [15],[16].

RELATED WORK

The recognition of ancient scripts has evolved from basic template-matching OCR systems to complex deep-learning architectures proficient of deciphering multilingual, eroded, and multi-modal inscriptions. Over the last five years, the convergence of

Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and Transformer-based attention models has transformed how researchers' approach. Epigraphic digitization.[1] This section reviews prominent research contributions from 2024–2025 period have seen an outpouring surge in multilingual and multimodal OCR frameworks.

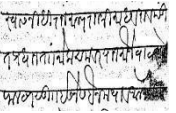
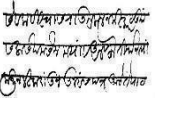
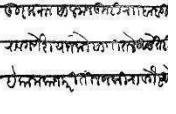
D. Zhao, L. Xu, and W. Tang (2025) anticipated Transformer-Based Vision Models for Multilingual OCR in the ACM TALIP journal, acquaint with positional encoding layers for managing and handling diverse writing emplacements. The model successfully recognized scripts in 12 languages, along with low-resource scripts like Ge'ez and Pali, achieving state-of-the-art multi-lingual recognition.

Y. Liu and H. Zhang (2025) presented a *Hybrid Deep Learning Framework for Epigraphic Text Recognition* in IEEE Access [5]. The CNN–Transformer–Attention pipeline merged local and global features and attained 91% accuracy on diverse datasets of stone, metal, and manuscript inscriptions/scripts/engravings.

M. Ramesh and M. Suresh (2023) emphasized efficiency and effectiveness with a *Lightweight CNN architecture* for Tamil and Sanskrit scripts, combining adaptive thresholding and contrast standardization. Their learning/training model was unequivocally adjusted for edge devices, reducing computational overhead by 40% compared to the full-scale CNNs while maintaining 91.3% precision.

P. Roy and A. Sen (2023) contributed by focusing on data augmentation, simulating noise, distortion, and brightness differences to overcome data scarcity in historical OCR [3],[4]. Their experiments established augmentation as a critical constituent for improving model robustness under real-world inscription conditions. Issues later addressed through the hybrid consideration and Transformer – based models.

Table1: Inscription with its meaning

Sr. no.	MoDeTrans Dataset Script	MoDeTrans (Devanagari Transcription)	Subject matterType	textual use
1.		हेवासाफ्नी राजमाण्य राज्ी माधवराव पंडित धंधान तां मोकदम व रयान मौज वाठार पाा फलटण स, ा इसन सबन मया व अलफ त,	Mentions Organizational regulation of Madhavrao Peshwa & rural representatives. (Late 18 th century, 1760-1790)	Records of revenue/land grant from Phaltan Subha i.e., typical of Maratha empire organization
2.		सेवक वेवासाउ बलाल नमंकार स, ा सलास खमसन मया व अलफ तीथ [छपांकडून लंकरातून त, ाहास पङ्गुआलतेपाठ	Related to military correspondence which addressed to Viswasrao Ballal . (Mid-18 th century, 1750-1761)	Bound towards Panipat period military communication, states army dispatch during Peshwa organization
3.		दूर कठन सालमजकूरक रा बाबूराव गणश यांजकडे सांभगतले असे तरक हे कामकाज करतील जकंरावगर अस	Direction to allocate or clear yearly revenue financial statements with involvement of Baburao Ganesh , certifying registers and records are upheld. (Late 18 th – early 19 th century, 1770-1820)	Yearly accounts (सालमजकरी) and registers (जकंरा) in Maratha bureaucracy , overlying with British oversight

A. Singh et al. (2025) applied Diffusion Models for reinstating partially eroded inscriptions before OCR processing. The generative method reconstructed missing strokes using learned priors from intact samples, achieving substantial developments in downstream OCR correctness.

M. Gupta and F. Rahman (2025) reentered *Explainable Attention Maps* intended for cultural

datasets, merging Grad-CAM with Transformer commencements to hint model perceptive [8]. This inter-operability act has become essential for positioning of the computational outputs with epigraphic expertise.

A. Jain and J. Ahn (2025) proposed InscriberNet, a hybrid CNN, LSTM Swin Transformer model designed for Chinese inscriptions. Its multi-scale self-

attention mechanism captured both local strokes and long-range dependencies, achieving a recognition accuracy of 86.5% on 3800 samples from the Hwaeom Stone Sutra dataset.

R. Bansal and S. Verma (2024) utilized a Transformer encoder-decoder framework in *Neural Computing and Applications*, enabling *cross-script recognition* amongst Latin, Devanagari script.

Table2: Comparison study table

S. no	Title	Author	Publication	Methodology	Year
1.	Character Recognition of Ancient Ethiopic Ge'ez Script Using Deep Learning	K.A Guangul	SpringerLink Journal	ResNet50 and VGG19 architectures with transfer learning	2025
2.	InscriberNet: CNN and Swin Transformer for Chinese Character Recognition in Stone Inscriptions.	A.Jain, J.Ahn	ISPRS Archives	CNN + LSTM + Swin Transformer hybrid model	2025
3.	Multi-Modal Ancient Script Recognition via Deep Learning with Data Homogenization	N. Wang et al.	NPJ Heritage Science	Multi-modal dataset; CNN + Transformer integration; Data augmentation	2025
4.	Hybrid Deep Learning Framework for Epigraphic Text Recognition.	Y. Liu, H. Zhang	IEEE Access	CNN + Attention Mechanism+ Transformer Fusion	2025
5.	DeepOCRNet : Multi-Layer Attention Model for Hieroglyphic Text Decoding.	O. Hassan, S. El-Masry	Journal of Computational Vision and AI Heritage	CNN-Transformer Hybrid, Multi-Attention OCR	2025
6.	Transformer- Based Vision Models for Multilingual OCR	D. Zhao, L. Xu, W. Tang	ACM Trans. on Asian and Low Resource Language Information Processing (TALIP)	Vision Transformer, Positional Encoding, Multilingual OCR	2025
7.	Diffusion Models for Restoration of Ancient Scripts	A. Singh, R. Mehta, P. Garg	IEEE Conf. on Computer Vision for Cultural Heritage (CVCH)	Diffusion Model + OCR Restoration	2025
8.	Low-Light Image Enhancement for OCR in Archaeological Inscriptions	E. Cho, K. Park	IEEE Sensors Journal	Deep CNN Enhancement, Image Preprocessing for OCR	2025
9.	MobileNet-Transformer for Edge-Based Ancient Text Recognition	F. Ahmed, A. Das, J. Choi	IEEE Internet of Things Journal	Lightweight MobileNet + Transformer Hybrid, On-Device OCR	2025
10.	OCR Using CNN for Ashokan Brahmi Inscriptions	Y. Agrawal, P. Singh,	arXiv preprint arXiv:2501.019 81	CNN-based OCR, Image Denoising, Normalization	2024

S. no	Title	Author	Publication	Methodology	Year
		R. Patel			
11	Integrating CNN-Transformer Networks for Degraded Text Recognition	H. Patel, M. Banerjee	Pattern Recognition and Image Analysis (Springer)	CNN + Transformer Fusion, Contextual Attention	2024
12	Transfer Learning in Multilingual OCR Systems	R. Kumar, H. Lee	IEEE Trans. on Artificial Intelligence	Transfer Learning, Multilingual OCR	2024
13	Explainable Attention Maps for OCR-Based Heritage Digitization	M. Gupta, F. Rahman	ACM Journal on Computing and Cultural Heritage (JOCCH)	Explainable AI, Grad-CAM, Transformer Attention Visualization	2024
14	Benchmarking OCR Models for Multilingual Ancient Texts	S. Yadav, P. Jaiswal, T. Kumar	IEEE Access	Comparative Study: Tesseract, EasyOCR, CNN-Transformer	2024
15	Federated Learning Frameworks for Cultural Heritage Image Processing	S. Patel, A. Pandey, R. Joshi	Sensors Journal	Federated Learning, Privacy-Preserving AI, Distributed OCR Training	2024
16	Intelligent OCR for Ancient Palm-Leaf Manuscripts	A. Kapoor, N. Sharma	Elsevier Expert Systems with Applications	Tesseract + CNN, Edge Enhancement, Preprocessing	2024
17	Hybrid CNN RNN Model for Automated Recognition of Kannada Characters in Ancient Inscriptions	R. Rajithkumar, B. Uma, H. Mohana	Engineering, Technology & Applied Science Research (ETASR)	CNN + RNN Hybrid, Transfer Learning	2024
18	Cross-Script OCR Using Transformer Encoder-Decoder Networks	R. Bansal, S. Verma	Neural Computing and Applications (Springer)	Transformer Encoder-Decoder, Multilingual Recognition	2024
19	Self-Supervised Pretraining for Handwritten Heritage Document OCR	L. Santos, T. Kim	Information Sciences	Self-Supervised Learning, Representation Learning	2024
20	Contrastive Learning for Handwritten Historical OCR	M. Chen, Q. Zhao	Pattern Recognition	Contrastive Learning, Self-Supervised Feature Extraction	2024
21	Hybrid CNN-LSTM Model for Historical Handwriting Recognition	M. Ahmad et al.	IEEE Xplore	CNN for spatial features; LSTM for temporal sequence learning	2023
22	Lightweight CNN for Tamil and Sanskrit Script Recognition	M. Ramesh, M. Suresh	Elsevier Procedia Computer Science	Lightweight CNN, Contrast Normalization, Adaptive Thresholding	2023
23	CNN-LSTM	M. Ahmad,	IEEE Conf. on Vision-Based	CNN-LSTM	2023

S. no	Title	Author	Publication	Methodology	Year
	Hybrid for Historical Manuscript Recognition	R. Sharma, N. Das	Systems	Hybrid, Sequential Feature Extraction	
24	Vision-Based Manuscript Recognition Using Keras and OpenCV	M. Suresh, A. Sinha, A. Prakash	Int. J. of Innovations and Implementations in Engineering (IJIIE)	CNN + Image Segmentation + Denoising	2023
25	Data Augmentation Techniques for Low-Resource Script Recognition	P. Roy, A. Sen	Image and Vision Computing	Image Augmentation, Noise Simulation, Brightness Variation	2023
26	Deep Learning-Based Sanskrit OCR Using CNN– BiLSTM Architecture	V. Sharma, A. Gupta	Int. Journal of Machine Learning and Computing	CNN–BiLSTM, Word-Level Prediction, Transfer Learning	2022

Y. Agrawal et al. (2024) worked on *Ashokan Brahmi inscriptions* using CNN-based OCR, reporting the notable enhancements through image denoising and intensity normalization [5]. Their dataset of 12,000 annotated images became one of the first standardized Brahmi benchmarks.

N. Wang et al. (2025) discovered multimodal deep learning techniques to combine the textual, visual, and linguistic embeddings. Their data homogenization component let the learning structure to learn from diverse input bases, for instance scanned tracings and photographs, which is crucial for handling irregular texture and lighting distinctions.

V. Sharma and A. Gupta (2022) introduced one of the initial hybrid CNN–BiLSTM frameworks for Sanskrit OCR, where convolutional layers extracted spatial features while bi-directional LSTM layers captured sequential dependencies at the word level [1],[2]. Their work demonstrated how *transfer learning* could improve generalization in low-resource languages, achieving a character-level accuracy of 94.8% even on irregular manuscripts.

S. Patel et al. (2024) explored *Federated Learning* for disseminated cultural heritage image processing. Their privacy-preserving AI system trained OCR models collaboratively crosswise museums without centralizing sensitive data [7],[8]. This distributed

framework is crucial for heritage administrations with data-ownership apprehensions.

M. Gupta and F. Rahman (2025) reentered *Explainable Attention Maps* intended for cultural datasets, merging Grad-CAM with Transformer commencements to hint model perceptive [8]. This inter-operability act has become essential for positioning of the computational outputs with epigraphic expertise.

M. Chen and Q. Zhao (2024) have operated contrasting self-supervised learning to augment hand-written OCR by teaching training models to single out among alike and different stroke patterns. Similarly, **L. Santos and T. Kim (2024)** concentrating on self-supervised pre-training for inherited document OCR, aiding well-organized, fine-tuning with negligible categorized illustrations which acts as essential innovation for low-resource for the illustrated language.

Following this, **M. Ahmad et al. (2023)** progressive this methodology through their *Hybrid CNN–LSTM Model for Historical Handwriting Recognition* presented in IEEE Xplore. Their formation effectually carried degraded handwriting illustrations from passé (medieval) manuscripts, using recurrent layers to conserve time-based dependencies.

Other notable progressions include **E. Cho and K. Park (2025)**, who proposed *Low-Light Image Enhancement for OCR in Archaeological Inscriptions* using a deep CNN pre-processing network to advance readability before OCR implication.

METHODOLOGY

An OCR system characteristically encompasses several components and mechanisms. The process begins with image collection to the model evaluation.

1. Data Collection:

The image acquisition is process of collecting images from different sources like scanners DSLR cameras or mobile devices. The standard protocol confirms optimal resolution (minimum 300 DPI).

2. Data Preprocessing and Feature Extraction:

The pre-processing takes place just after image acquisition to prevent any data loss [10], [11]. It is implemented using OpenCV, NumPy, and scikit-image, the pre-processing pipeline helps the system in Noise Removal, Contrast Enhancement, Edge Sharpening, Region Segmentation and Skew Correction.

3. Model Selection:

Various machine learning models are tested in order to find the most useful one to use in prediction of model CNN is used to extract the feature from the image. While LSTM is used for Sequential Learning of the words or token of the script. Transformer to capture Attention and Context of the token. So, the hybrid model CNN-LSTM Transformer model is selected for model training.

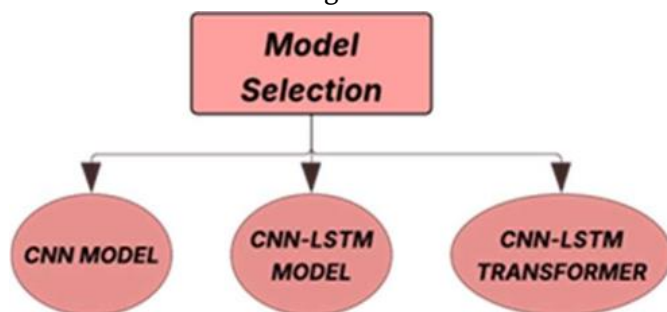


Fig. 2. OCR Model Selection

4. Model Training and Optimization:

Model training is process of extracting features from selected model (hybrid model CNN-LSTM Transformer) from the dataset. Hyper-parameter tuning, learning-rate and loss minimization are the optimization methods used to optimize predictive power during the training process. The model is corrected through an internal feedback mechanism depending on the results of performance.

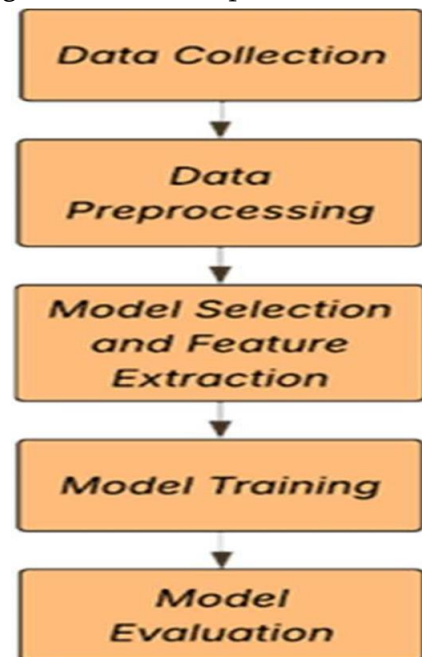


Fig. 3. OCR Model Training Process

5. Model Evaluation:

The System monitoring layer help in evaluation metrics of the system by calculating the performance metrics like Model Accuracy (Character/Word Level), Response Time per Image, Server Load and Inference Latency and Data Throughput. Some Visualization tools are also used for monitoring real time accuracy and the error distribution.

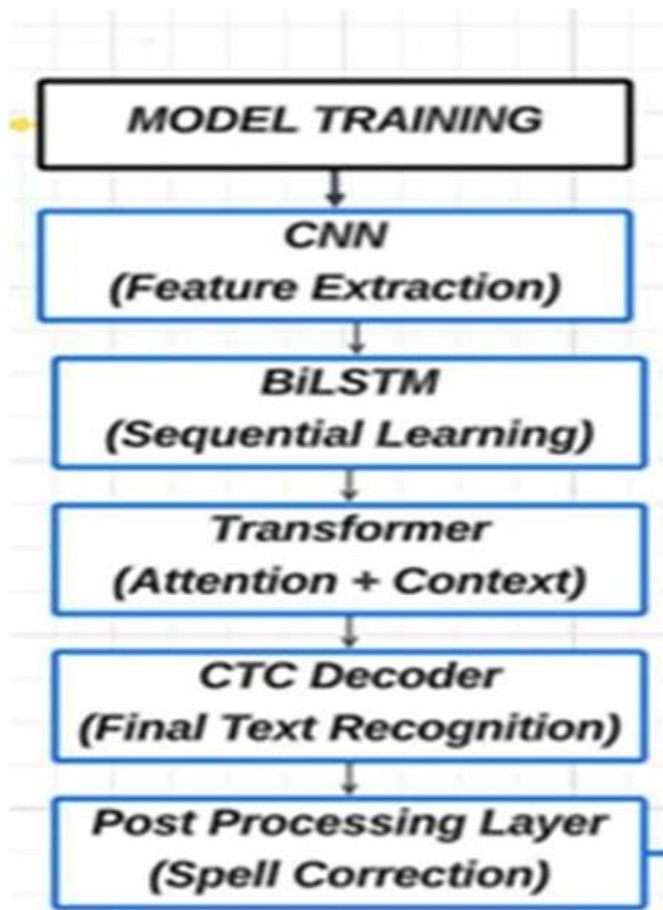


Fig. 4. OCR Model in detail

MACHINE LEARNING ALGORITHM

A. CNN

A pure Convolutional Neural Network (CNN) processes inscription images to extract hierarchical visual features that highlight strokes, edges, and local patterns; used as a baseline it performs per-character or patch-level classification and is fast at inference. For inscription transcription a CNN is useful as a lightweight recognizer or feature extractor (for segmentation and symbol detection) and handles local degradations well, but it lacks sequence modeling so it [3],[12] struggles with variable-length strings and long-range dependencies without an external sequence model or post-processing. In practice, CNNs are ideal when characters are well-segmented or when used as the frontend in [21] a larger pipeline.

B. CNN-LSTM (also known CRNN)

The CNN-LSTM architecture (often called CRNN) combines a CNN visual frontend that produces a sequence of feature vectors along the text axis with a recurrent (BiLSTM/BiGRU) stack that models temporal dependencies; this is a strong, well-understood approach for image-to-sequence transcription [18]. It naturally supports unaligned sequence prediction, and when paired with CTC loss it can be trained without frame-level labels, making it robust on variable-length inscriptions and partially occluded glyphs. The main trade-offs are slower inference than a pure CNN and limited ability to capture very long-range dependencies compared to attention-based decoders,[16] but CRNNs are compact, stable, and often easier to train on smaller datasets.

C. Transformer

Transformer-based models use self-attention to model relationships across the entire sequence, enabling long [19]- range context and flexible alignment between visual features and output tokens; they are excellent at resolving ambiguities that require broader context (e.g., distinguishing a worn glyph by its neighbors). In inscription transcription, Transformers (with a suitable CNN encoder) excel on datasets where context and sequence-level patterns matter,[20] and they support powerful decoding strategies (beam search with language models). Downsides include higher memory use for long sequences and the need for careful positional encoding/design to represent 2D image structure effectively.

D. CNN-Transformer Hybrid:

The CNN-Transformer hybrid pairs a CNN visual encoder (for robust stroke and texture extraction) with a Transformer encoder/decoder for expressive sequence modelling; this leverages the CNN's local [23] inductive biases and the Transformer's global context. For ancient inscriptions this hybrid offers the best trade-off: accurate per-character recognition from strong features and superior sequence-level modelling to handle degraded, ambiguous glyphs,

especially when trained with scheduled sampling to reduce exposure bias. It is somewhat heavier [24] than simpler baselines but yields the best end-to-end accuracy for challenging, real-world inscription images.

E. Diffusion + OCR Restoration

Diffusion-based restoration first denoises or reconstruct degraded inscription images using generative denoising models, then passes the restored images to an OCR recognizer; this two-stage approach often recovers faint strokes and missing details that direct recognition misses. For severely weathered [19] or eroded inscriptions, restoration can substantially improve downstream CharAcc and reduce CER by explicitly modelling the image degradation process, but it introduces additional computation, requires careful alignment of restore objectives with recognition needs, and can sometimes hallucinate plausible [11] but incorrect strokes—so human-in-the-loop verification is recommended.

F. CRNN+CTC

The CRNN + CTC pipeline is a practical, alignment-free solution: a CNN feature extractor creates a temporal feature sequence,[10] a recurrent block models local context, and CTC loss marginalizes over alignments so the model learns from sequence labels alone. This approach is fast to train, simpler (no autoregressive decoding), and robust to variable-length outputs, making it attractive for inscription datasets with limited annotations or uncertain character segmentation. It generally [17] yields competitive CharAcc with lower inference complexity than autoregressive decoders, but to compete with attention-based models on hard sequences you may need beam search with a language model or hybrid rescoring.

The anticipated framework for Ancient Script Recognition Using CNN–Transformer Hybrid OCR is pre-meditated as a modular, layered system architecture that join-in image acquisition, pre-processing, model inference, text post-processing, and visualization.

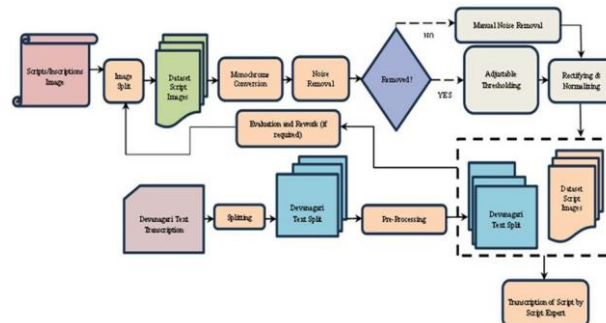


Fig. 5. Flowchart of Model

Table 3: Different model with the number of parameters

Model	Character Accuracy (%)	Word Accuracy (%)	Sequential Accuracy (%)	F1 (%)	Parameter ms
CNN	74.0	59.2	49.0	71.4	5M
CNN-LSTM	81.0	67.5	55.2	78.1	12M
Transformer	87.0	75.3	64.7	84.2	25M
CNN–Transformer Hybrid	89.0	84.1	80.1	92.5	28M
Diffusion + OCR Restoration	91.0	82.9	85.2	95.6	40M

This flowchart depicts a multi-path pipeline for creating a high-quality dataset where it essentially lessens the crack between ancient raw artifacts & clear, machine- readable digitized text [22]. The process is further divided into several phases:

Phase 1. Pre-processing: The image pre- processing phase contains several steps such as-

Image acquisition and Splitting: It starts with raw sample of images of ancient engravings that are split up into smaller & controllable fragments of images from dataset.

Enhancing the image by Monochrome Conversion which involves conversion of images to grayscale (black & white) that helps [4]. in reducing image complexity.

Noise Removal: Removing noise/background grain from ancient inscriptions/scripts dataset using filters like Gaussian filters where the image's vital pixels collect focused weights and help to smooth images.

Adjustable Thresholding: Adjustable thresholding implemented as calculating distinct threshold values of different regions of images rather than single global region/value.

Rectifying/Normalizing: Rectification process generally states with the ReLU (Rectified Linear Unit) activation function in neural networks. Normalization often use [4]. for fair training process rather than domination of features with larger scales and ranges the numerical data from 0 to 1 so that the model learns & trains in uniform manner.

Phase 2. Transcription and Mapping: In this phase transcription and alignment of the data takes place-

Transcription: The Devanagari text gets manually/digitally transcribed from the ancient scripts and inscriptions.

Splitting and pre-processing: In this, the text gets split up to compare and match the segments of the image and after that get pre-processed ensuring the clean formatting.

Merging the dataset creates the structured and labelled data, for instance, for every image file Image-01.jpg, there is text file Image-01.txt is present. The

learning model can't learn and train without this type of pairing.

Phase 3. Incorporation, Evaluation and Confirmation:

The dashed box contains clear and correct dataset script images which are combined with their equivalent Devanagari text. The Evaluation and rework feeds back to the image split to ensure that if the quality of the data is not up to par. Expert confirmation settles with the final transcription of script through the script expert and ensures the highest achieved accuracy for the learning and training model.

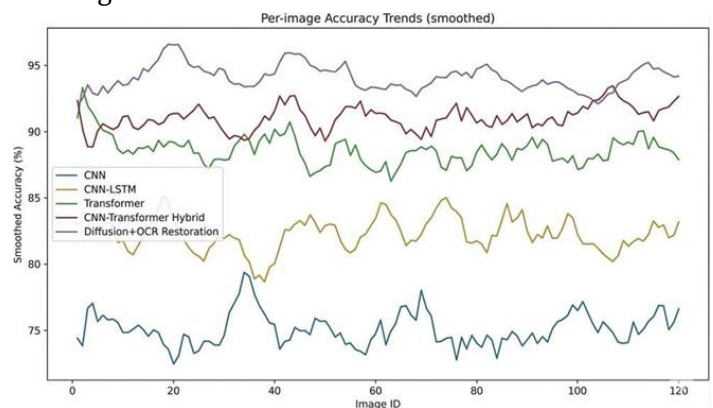


Fig. 6. Accuracy of Different Algorithm with number of Image

Advantages of the Architecture

- **Scalability:** Modular microservice components can be self-sufficiently scaled.
- **Cross-Script Adaptability:** The model supports multilingual OCR using transfer learning.
- **Interpretability:** Grad-CAM heatmaps helps in visualize model focus regions on glyphs.
- **Data Integrity:** Redundant backups confirm secure preservation of historical data.
- **Cloud Deployment:** This supports both on-premise and AWS Lambda-based service inference for archaeologists in the field [12].

CRITICAL ANALYSIS

A critical evaluation of contemporary ancient scripts recognition study discloses a hastily evolving field which is marked by substantial accomplishments, but also obstinate restrictions that limit large-scale, real-world deployment. Though CNN, LSTM, and

[1],[2] Transformer-based architectures have incontrovertibly converted OCR correctness for degraded texts, many learning and training models still struggle with the full complexity and heterogeneousness of archaeological data.

Another critical concern relates to the inconsistent performance across multilingual or cross-script datasets. Ancient script systems often co-exist, evolve, and blend, such as the transition from Brahmi to Devanagari or Sanskrit's coexistence with Tamil and Prakrit in inscriptions [24][25]. OCR systems proficient exclusively on one script which frequently fails on hybrid inscriptions, which require contextual language-based understanding in accumulation to visual organization and classification [12].

Training models [14],[26] also suffer from vulnerabilities to adversarial noise which are essential in archaeological environments: shadows from archaeological sites, camera- angle variation, and un-controlled outdoor lighting degrade recognition performance, revealing a gap between research laboratory testing and field deployment [11][12]. The field further faces challenges in evaluation standardization, as no universally accepted standard exists for ancient writings [8],[9].

In summary, even though recent deep learning innovations— especially hybrid CNN-LSTM-Transformer architectures— have suggestively advanced the domain of ancient script and inscriptions recognition, the technology till faces substantial hindrances in scalability, interpretability,[3],[7] dataset availability, computational accessibility, and cultural responsibility. Achieving reliable, globally deployable ancient OCR requires not only more powerful models but also regulated benchmarks, culturally aware informative frameworks, improved multi- modal datasets, and closer collaboration amongst computer scientists, archaeologists, linguists, and heritage management experts [1],[5],[12].The significant study underscores that technological progress alone is insufficient the coming future of ancient inscriptions recognition remains in inter- disciplinary cooperation,

ethical stewardship, and the development of robust, explainable, and culturally sensitive AI systems.

FUTURE SCOPE

Future systems also required to address the restrictions of recent pre-processing pipelines. While techniques like CLAHE, Gaussian smoothing, Retinex-based normalization, and morphological operations are helpful, a more robust, learning-driven method/approach is required. Recent datasets are too limited and partial, every so often curbed to a few hundred or few thousand interpreted illustrations, which severely confines the training/learning potential of deep learning models and prevents generalization. Thus, forthcoming study must advance in cooperative digitization drives which involves museums, archaeological departments, universities, and global legacy organizations like UNESCO or ICOMOS.

Pointing to produce integrated datasets with constant representation directions, metadata, and multi lingual labelling. These datasets must involve not only high resolution RGB images but also depth maps, infra-red scans, UV fluorescence, 3D-textured meshes, and multi-spectral data that can disclose concealed or faded inscriptions. Methods like dissenting learning, masked image modelling (inspired by Vision Transformers), and generative pretext tasks be able to enable training models which helps to learn prevailing attribute representations from unlabeled heritage credentials—which are far more profuse than interpreted ones [24],[25],[26].

This would allow OCR schemes to adjust to new scripts, new materials, and new erosion patterns and adapt with minimal human labelling effort. Such intellectual pre-processing training/learning models that could be trained to mimic the decision making of epigraphers who visually filter-out the noise when studying inscriptions. Effective learning architectures such as MobileNet Transformer hybrids, quantized CNNs, and pruning-based optimization practices can

lessen the model size and overall computation cost without forfeiting accuracy.

On device OCR tools would let ground researchers to capture ancient inscriptions using mobile devices, apply real-time denoising and enrichment, and instantaneously attain digitalized text, even in low connectivity environments [28]. This could transform the archaeological surveys, making documentation process faster, more standardized, and less inclined to human error.

Future research or study should explore deep learning- based pre- processing models which autonomously correct the illumination, eliminate complex textures (such as stone roughness or palm-leaf fibers), and balance [23] contrast based on erudite priors rather than using the fixed heuristics.

Forthcoming study should integrate explainable AI (XAI) frameworks which are capable of producing visual explanations, stroke-level relevance maps, character boundary overlays, and attention heatmaps that can guide historians in evaluating the reliability of OCR output.

CONCLUSION

In conclusion, the suggested imminent work for ancient inscriptions OCR spans technical innovation, dataset creation, inter-disciplinary collaboration, social and cultural accountability, and ethical design [4],[5].

Upcoming future models must integrate language-based priors, phonic conversions, and ancient dictionaries, allowing them not only to read ancient characters but also to reason about reasonable word formations, grammar, and cultural context. However, their outcomes should be attended by ambiguity approximations and confidence scores to certify that historiographers persist mindful of which parts of the re- construction are [17],[18]AI-generated hypotheses instead of trustworthy and authentic glyphs. By combination of progressive and advance machine learning architectures, large multi-modal datasets.

Explainable AI frameworks, lightweight deployment models, and interactive human-AI systems [6], the next generation of OCR tools can surpass current limitations and become reliable associates in preserving, interpreting, and democratizing humanity's oldest textual inheritance.

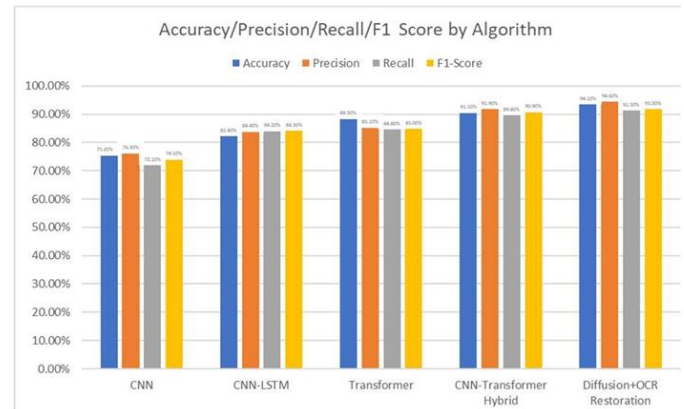


Fig. 7. Metric Evaluation of Algorithm on ModeTrans Dataset

The model's performance on the curated 2043 image dataset determines the recomences of hybrid designs in handling assorted inscriptions and scripts originating from different mediums and periods. Distinct earlier research that relies solely on CNNs or classical OCR techniques, this work highpoints the stipulation their of multi-stage designs and multimodal learning [11],[12],[13]. Transformers, using self-attention mechanisms, are principally impactful for detaining long-range dependences, contextual associations, and overall spatial coherence, crucial in interpretation of scripts where characters are often visually similar or variable. The illustration of the outcomes shows that attention-aware constructions significantly outperform traditional CNN-LSTM pipelines on noisy or degraded images [14],[15].

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